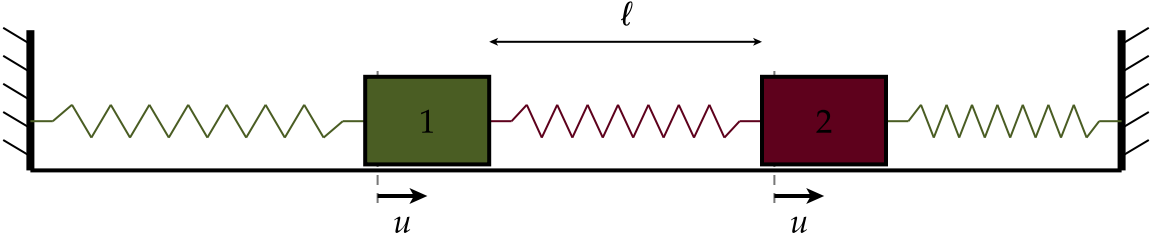
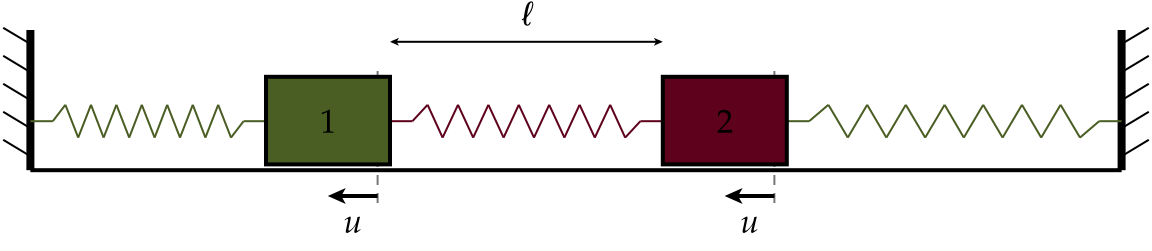
(a) Both carts pushed right by the same amount: $u_1 = u_2 = u$



(b) Half a period later: both carts left by the same amount